

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 640 Quantum Algorithms — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Nielsen & Chuang — Quantum Computation and Quantum Information	The primary text, and the standard one. Modules 02 through 10 follow its development. Library copy; it is demanding and it is worth the demand.
Aaronson — Quantum Computing Since Democritus, and his lecture notes (free)	Modules 01, 12, and 13. The best available correction to the popular account, from somebody who works in the field and says plainly what is not known.
Kitaev, Shen & Vyalıy — Classical and Quantum Computation	Modules 07, 08, and 12. The complexity-theoretic treatment, and the phase estimation development is especially clean. Library copy.
Preskill's lecture notes on quantum computation (free)	Modules 04, 10, and 11, free in full. The error correction and threshold material is the clearest written account, by the person who named the current era.
The Qiskit textbook and documentation (free)	The implementation side, Modules 03 onward — and simulating a circuit you derived on paper is the fastest way to find out you derived it wrong.
Gidney & Ekerå — How to factor 2048 bit RSA integers (free)	Modules 10 and 13's resource counts. A careful, specific estimate of what Shor's algorithm would actually cost in physical qubits and hours, which is the number to quote.

Module-by-module map

Module 01 — What a Quantum Computer Is

Resource	What it covers for this module
Aaronson — The Limits of Quantum Computers, and related essays (free)	§§2 and 3 — the misconceptions corrected by somebody in the field, and the 'no, it does not try all answers' argument in its original form.
Nielsen & Chuang, chapter 1 (library copy)	§1 — the model stated carefully, and the historical development of why it was proposed.
Feynman — Simulating Physics with Computers (free copies widely available)	§3's first item in the original — the proposal that started the subject, and it is about simulation rather than about speed.
Preskill — Quantum Computing in the NISQ era and beyond (free)	§§2 and 4 — the honest assessment of where the hardware is, from 2018 and still the right framing.

Module 02 — States and Measurement

Resource	What it covers for this module
Nielsen & Chuang, chapters 2 and 4 (library copy)	§§1 to 3 — the linear algebra and the measurement postulates, developed carefully.
Wootters & Zurek — A single quantum cannot be cloned	§3's callout in the original — one page, and the argument is the one you can reconstruct.
Watrous's notes, the early lectures (free)	§§1 and 2, rigorous and short — and the measurement formalism is stated more generally than Nielsen-Chuang does at first.
Aaronson — Multilinear formulas and the Gottesman-Knill theorem, lecture notes (free)	§4's qualification — which circuit classes simulate classically, and why that matters for advantage claims.

Module 03 — Gates and Circuits

Resource	What it covers for this module
Nielsen & Chuang, chapter 4 (library copy)	§§1 and 2 — the gate set, universality, and the circuit constructions, with the Solovay-Kitaev discussion.
Bennett — Logical reversibility of computation	§2's uncomputation trick in the original , and it predates quantum computing entirely.
Amy et al. — T-count optimisation (free preprints)	§4's metric — why T-count is the target, and how it is reduced in practice.
The Qiskit transpiler documentation (free)	§4 concretely — run your circuit through it and watch the gate count grow as connectivity constraints are applied.

Module 04 — Interference

Resource	What it covers for this module
Preskill's notes, the lectures on quantum information (free)	§§2 and 3 — decoherence and which-path information, developed properly rather than by analogy.
Nielsen & Chuang, chapters 6 and 8 (library copy)	§1's path view and §3's open-system formalism.
Zurek — Decoherence and the transition from quantum to classical (free)	§§2 and 3 — the which-path account of decoherence from the person who developed it.
Aaronson — lecture notes on the source of quantum speedups (free)	§§1 and 4 — and the 'what cancels' question treated as the central one, which it is.

Module 05 — The Query Model

Resource	What it covers for this module
Deutsch & Jozsa — Rapid solution of problems by quantum computation	§2 in the original — short, and the promise structure is stated clearly.
Simon — On the power of quantum computation	§3 in the original — and the paper Shor credits as the one that suggested the approach.
Watrous's notes on query complexity (free)	§§1 and 4 — the model stated rigorously, with the lower-bound methods.
Beals et al. — Quantum lower bounds by polynomials (free)	§4's technique — the polynomial method, which is how most query lower bounds including Grover's are proved.

Module 06 — Grover's Algorithm

Resource	What it covers for this module
Grover — A fast quantum mechanical algorithm for database search (free)	§§1 and 2 in the original — four pages, and the geometric picture is not the one he used, which is instructive.
Bennett, Bernstein, Brassard & Vazirani — Strengths and weaknesses of quantum computing (free)	§3's lower bound — and it is the paper that bounds the whole subject's search-based hopes.
Brassard, Høyer, Mosca & Tapp — Quantum amplitude amplification and estimation (free)	§2's generalisation — the template, stated in full generality.
Babbush et al. — Focus beyond quadratic speedups (free)	§4's assessment, from people building the hardware — and the conclusion is the honest one.

Module 07 — The Quantum Fourier Transform

Resource	What it covers for this module
Nielsen & Chuang, chapter 5 (library copy)	§§1 and 2 — the QFT, its circuit, and phase estimation, in the standard development.
Coppersmith — An approximate Fourier transform useful in quantum factoring (free)	§2's approximate version in the original — and the argument for dropping the small rotations.
Kitaev, Shen & Vyalı, the phase estimation chapter	§1 and Module 08 — the cleanest derivation, and the one that makes the eigenvalue framing central. Library copy.
Aaronson — Read the fine print (free)	§3's input problem and non-uses — the standard reference for what the caveats actually cost, and it should be read before any quantum machine learning proposal.

Module 08 — Phase Estimation and Shor

Resource	What it covers for this module
Shor — Polynomial-time algorithms for prime factorization and discrete logarithms (free)	§§1 and 2 in the original — and it is readable, which is not true of every landmark paper.
Kitaev — Quantum measurements and the abelian stabilizer problem (free)	§1 — phase estimation in the framing that makes it the general tool rather than a step in Shor.
Gidney & Ekerå — How to factor 2048 bit RSA integers in 8 hours (free)	§4's numbers — the careful estimate, with every assumption stated. This is the paper to cite.
NIST post-quantum cryptography standards (free)	§3's replacements — the standardised algorithms and the migration guidance.

Module 09 — Quantum Simulation

Resource	What it covers for this module
Feynman — Simulating Physics with Computers	§1 in the original — the proposal, and it is worth reading for how carefully he states the difficulty.
Lloyd — Universal quantum simulators	§2 — the Trotterisation argument that turned the proposal into an algorithm.
Low & Chuang — Hamiltonian simulation by qubitization (free)	§2's modern method — near-optimal, and what current resource estimates assume.
Reiher et al. — Elucidating reaction mechanisms on quantum computers (free)	§§3 and 4 — a concrete application with a full resource estimate, which is the model for how this should be reported.

Module 10 — Noise and Error Correction

Resource	What it covers for this module
Shor — Scheme for reducing decoherence in quantum memory	§§1 and 2 — the first quantum code, and the phase-flip problem solved.
Fowler et al. — Surface codes: towards practical large-scale quantum computation (free)	§§2 to 4 — the standard reference, with the overhead numbers and the decoding problem.
Preskill — Fault-tolerant quantum computation (free)	§3's threshold theorem , explained rather than only proved.
Google Quantum AI — below-threshold surface code results (free)	§§3 and 4 experimentally — a logical qubit outperforming its physical parts, which is the milestone to understand.

Module 11 — Near-Term Hardware

Resource	What it covers for this module
Preskill — Quantum Computing in the NISQ era and beyond (free)	§1 — the framing, from the person who named it, and it is admirably careful about what is and is not claimed.
Cerezo et al. — Variational quantum algorithms (free)	§2 — the survey, with the obstacles stated alongside the hopes.
McClean et al. — Barren plateaus in quantum neural network training landscapes (free)	§3 in the original — and the result is cleaner and more damaging than its reception suggested.
Aaronson — Read the fine print (free)	§4's checklist — the input and output problems stated once and for all, and the right thing to hand somebody with a proposal.

Module 12 — BQP and Its Neighbours

Resource	What it covers for this module
Bernstein & Vazirani — Quantum complexity theory	§1 — where BQP is defined and its robustness established.
Aaronson — BQP and the Polynomial Hierarchy (free)	§2's containments — and the oracle evidence for BQP containing things outside NP.
The Complexity Zoo (free)	§2 — look up BQP and follow the containments, which is the fastest way to see the landscape.
Aaronson — The Limits of Quantum Computers (free)	§3 — the NP question treated carefully and for a general audience, which is harder than it sounds.

Module 13 — Claiming a Quantum Advantage

Resource	What it covers for this module
Aaronson — Read the fine print (free)	§§1 and 2 — the caveats, stated once and carefully, and the right thing to read before assessing any proposal.
Tang — A quantum-inspired classical algorithm for recommendation systems (free)	§2's dequantisation — the paper that started the pattern, and it is short.
Pan, Chen & Zhang — classical simulation of random circuit sampling (free preprints)	§2's second kind — a claimed classical cost reduced by orders of magnitude after the fact.
Quantum (the journal), the resource-estimate papers (free)	§§1 and 4 — the papers that do state all three things, which are the model to imitate in Project 2.